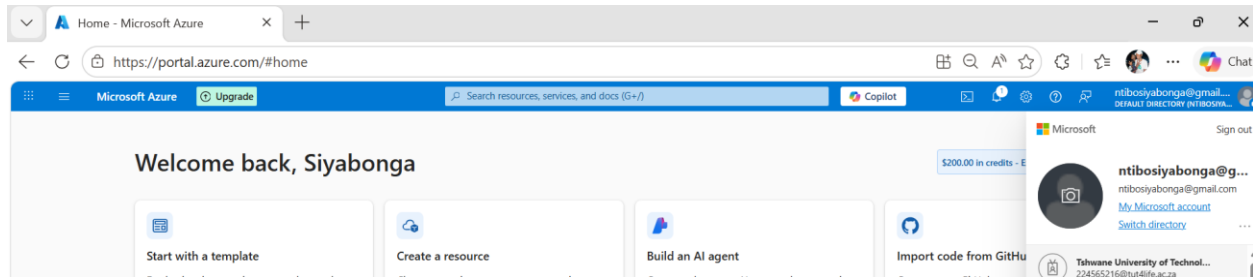


TASK 1: Authentication and Authorization in Azure

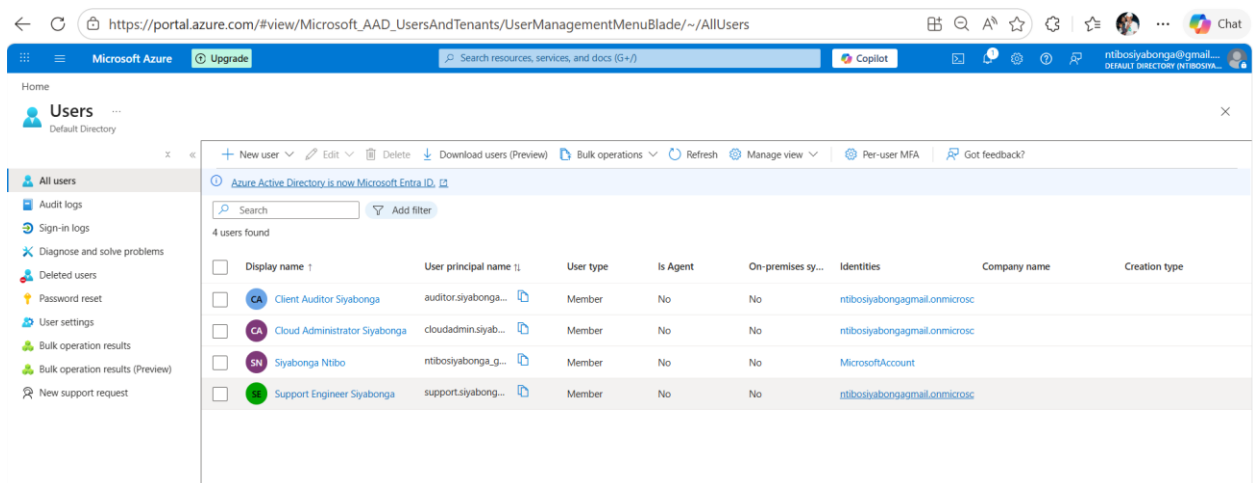
Step 1: Access Azure Identity Management

Go to <https://portal.azure.com> and log in with your account.



The objective of this task is to demonstrate the ability to configure identity and access management in Azure using Microsoft Entra ID, including users, groups, role-based access control (RBAC), Multi-Factor Authentication (MFA), and Conditional Access.

Step 2: Create Users



Step 3: Create Groups

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot ntibosiyabonga@gmail... ntibosiyabonga@gmail... DETAILS DIRECTORY (NTIBOSIYABONGA...)

Home

New Group

Got feedback?

Group type * ⓘ
Security

Group name * ⓘ
BCX-Operations-Siyabonga

Group description ⓘ
Group for BCX operations staff including Cloud Admins and Support Engineers

Membership type ⓘ
Assigned

Owners
No owners selected

Members
No members selected

Create

New Group

Got feedback?

Group type * ⓘ
Security

Group name * ⓘ
BCX-Auditors-Siyabonga

Group description ⓘ
Group for BCX client auditors with read-only access

Membership type ⓘ
Assigned

Owners
No owners selected

Members
No members selected

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Search resources, services, and docs (G+)

Home > Groups

Groups | All groups

Default Directory

Overview

All groups

Insights (Preview)

Deleted groups

Diagnose and solve problems

Settings

Activity

Troubleshooting + Support

Search

Add filter

Search mode: Contains

2 groups found

☐	Name ↑	Object Id	Group type	Membership type	Email
☐	BCX-Auditors-Siyabonga	ee4315d3-2b58-4d95-b9d0-ca7c6fbadaaf	Security	Assigned	
☐	BCX-Operations-Siyabonga	b0d3a889-5590-4d25-8a3b-75f15286ea36	Security	Assigned	

https://portal.azure.com/?feature.tokencaching=true&feature.internalgraphapiversion=true#view/Microsoft_AAD_IAM/GroupDetail...

Microsoft Azure | Upgrade

Search resources, services, and docs (G+)

Home > Groups > All groups > BCX-Auditors-Siyabonga

BCX-Auditors-Siyabonga | Owners

Group

Overview

Diagnose and solve problems

Manage

Properties

Members

Owners

Roles and administrators

Administrative units

Group memberships

Applications

Licenses

Azure role assignments

Activity

Troubleshooting + Support

Search

Add filter

1 group owner found

☐	Name ↑	Type	Email	User type	Object Id
☐	Client Auditor Siyabonga	User		Member	7cfa70d9-2453-4491-9515-142637bf7b10

Microsoft Azure | Upgrade

Search resources, services, and docs (G+)

Home > Groups > All groups > BCX-Operations-Siyabonga

BCX-Operations-Siyabonga | Owners

Group

Overview

Diagnose and solve problems

Manage

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Owners

Roles and administrators

Administrative units

Group memberships

Applications

Licenses

Azure role assignments

Activity

Troubleshooting + Support

Search

Add filter

2 group owners found

☐	Name ↑	Type	Email	User type	Object Id
☐	Cloud Administrator Siyabonga	User		Member	29ff0b33-9284-401e-a93d-660b57e6dcdd
☐	Support Engineer Siyabonga	User		Member	3b96502c-8c44-4ba6-8a83-a88d238d970a

Groups make it easier to manage permissions for multiple users at once.

Step 4: Configure Role-Based Access Control (RBAC)

Microsoft Azure

Upgrade

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Copilot

ntibosiyabonga@gmail...
DEFAULT DIRECTORY INTROSIVA...

Home > Resource Manager > Resource groups

Create a resource group ...

BasicsTagsReview + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription *

Azure subscription 1

Resource group name *

RG-BCX-Siyabonga

Region *

(Africa) South Africa North

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

ntibosiyabonga@gmail...
DEFAULT DIRECTORY INTROSIVA...

Home > RG-BCX-Siyabonga > Access control (IAM)

Add role assignment ...

RoleMembersConditionsReview + assign

Selected role

Owner

Assign access to

☒ User, group, or service principal
☐ Managed identity

Members

[+ Select members](#)

Name	Object ID	Type
BCX-Operations-Siyabonga	b0d3a889-5590-4d25-8a3b-79f15286ea...	Group

Description

Optional

Review + assign

Previous

Next

Feedback

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

ntibosiyabonga@gmail...
DEFAULT DIRECTORY INTROSIVA...

Home > RG-BCX-Siyabonga > Access control (IAM)

Add role assignment ...

RoleMembersConditionsReview + assign

Selected role

Reader

Assign access to

☒ User, group, or service principal
☐ Managed identity

Members

[+ Select members](#)

Name	Object ID	Type
BCX-Auditors-Siyabonga	ee4315d3-2b58-4d95-b9d0-ca7c6fbadaaf	Group

Description

Can view resources, but cannot modify anything

Review + assign

Previous

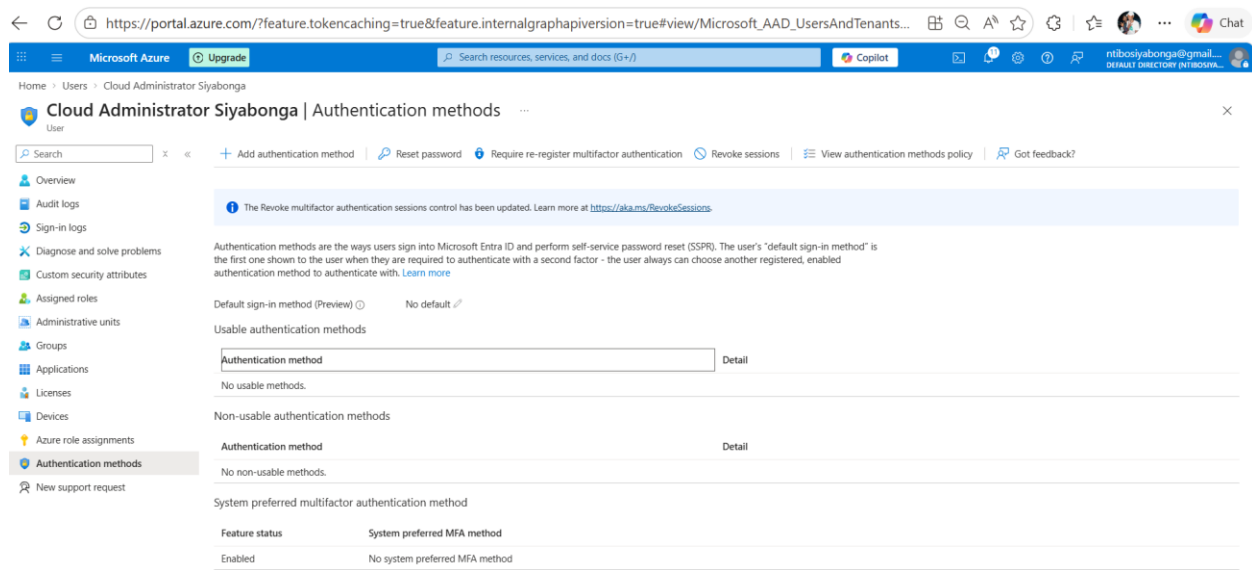
Next

Feedback

Authentication: Verifies who a user is.

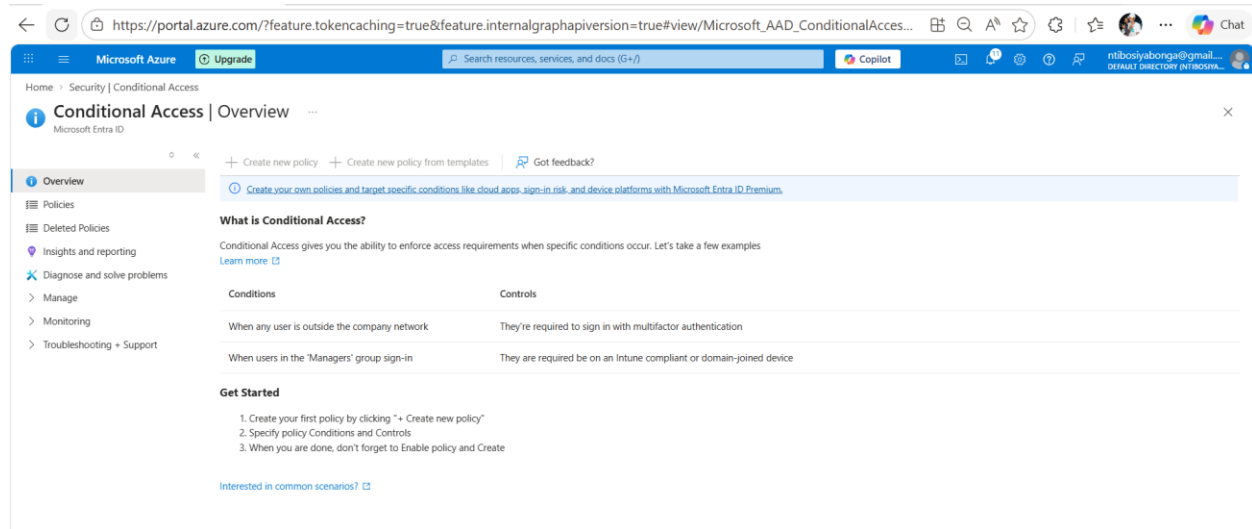
Authorization: Controls what an authenticated user is allowed to do. Authentication alone is not enough without proper RBAC.

Step 5: Enable Multi-Factor Authentication (MFA)



MFA adds a second factor, such as a mobile verification code, to secure accounts. Even if the password is stolen, unauthorized access is blocked.

Step 6: Configure Conditional Access



Conditional Access policies require Azure AD Premium. In a student & free tenant, the New Policy button is disabled. In a production environment, BCX would create a policy requiring MFA for all administrators, restricting access from untrusted locations, or enforcing compliant devices.

STEP 7: External Identities

BCX can use Azure External Identities (B2B collaboration) to allow third-party vendors or external auditors to securely access selected resources. External users are invited to the tenant and assigned limited roles using RBAC. This ensures that external access is controlled, monitored, and restricted to only necessary resources without exposing the entire environment.